

# Tristan David Carter

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## Education

**Duke University** Aug 2025 – May 2029  
B.S.E. Electrical & Computer Engineering (ML Concentration), B.S. Computer Science, GPA: 3.837/4.000  
**Relevant Coursework:** Linear Algebra, Data Structures & Algorithms, Computer Architecture, Signals & Systems

## Projects and Technical Experience

**Multimodal Time-Series Forecasting of Alzheimer's Disease Progression** Jan 2026 – Jul 2026

- Co-developed continuous-time models to forecast Alzheimer's progression using MRI, PET, and clinical data.
- Built a pipeline to organize MRI/PET scans by patient and visit; migrated datasets to AWS S3.
- Automated scan preprocessing: BIDS conversion, skull stripping, registration, quality control, and harmonization.
- Modeled changes in brain anatomy using optical flow and optimal transport flow matching.
- Forecast cognitive scores with a JAX/Diffrax Neural CDE; adding clinical inputs cut ADAS-13 validation MAE ~50%.

**Duke WATCH: AI Scraper Detection & LLM Attribution** — *Duke Code+ Intern* May 2026 – Aug 2026

- Built a FastAPI honeypot platform on AWS to track web scrapers and trace scraped content in LLM outputs.
- Automated Ansible deployment of Docker containers with Nginx routing and Prometheus/Grafana/Loki monitoring.
- Generated pages with LoRA-tuned Qwen3-1.7B; used RapidFuzz to match per-device canary tokens in LLM outputs.
- Tracked scrapers across IP changes using browser, behavioral, and Zeek JA4/JA4T signals stored in PostgreSQL.
- Flagged inconsistent scraper identities, clustered campaigns with Louvain, and inferred providers with GraphSAGE.

**ML Damping and Forcing Estimation in Spring–Mass Systems**, *2024 AIAA YPSE Conference* June – Nov 2024

- Trained DNN/LSTM models on 80,000 simulated trajectories to estimate damping and six forcing parameters.
- Validated RK4 data against Laplace solutions; tested models on 20,000 trajectories per scenario, including car struts.

**Optimizing Adaptive Cruise Control in Adverse Environments** Sept 2023 – Aug 2024

- Simulated nonlinear cruise control in Python with Lyapunov/sliding-mode control and Chebyshev II radar filters.
- Trained YOLOv5 on the Canadian Adverse Driving Conditions dataset for object detection in snow and rain.

**Full-Stack Database Management Application**, *National FBLA Conference 2024* Dec 2023 – July 2024

- Built a Java/MySQL web app on Apache Tomcat to store, search, and update business partner contacts.
- Implemented MVC with JSP/Servlets/JDBC; added WebSocket viewer tracking and snapshot restoration.

## Journal Papers

**Quasi Monte Carlo Simulations**, *Howard County Math Journal* ↗ Aug 2025

Compared Sobol and pseudorandom sampling for numerical integration, analyzing sample uniformity and convergence. Applied quasi-Monte Carlo to Asian option pricing under geometric Brownian motion.

**Parameter Estimation for a Linear System of Equations**, *Howard County Math Journal* ↗ Aug 2024

Derived least-squares and minimum-norm estimators using matrix calculus and Lagrange multipliers, minimizing residuals in overdetermined systems and parameter norms in underdetermined systems.

## Technical Stack

**Languages:** Python, C++, Java, JavaScript, SQL, Unix/Bash, HTML/CSS

**ML/AI Stack:** PyTorch, TensorFlow, JAX, scikit-learn, NumPy/SciPy/Pandas, Hugging Face, PEFT/LoRA, MuJoCo

**Backend/Database:** FastAPI, Flask, Uvicorn, REST APIs, Nginx, PostgreSQL, SQLite

**Infrastructure:** AWS (EC2, S3, IAM), Terraform, Ansible, Linux, Git, CI/CD, Prometheus, Grafana, Loki, PromQL

**Networking/Security:** TCP/IP, TLS/TCP fingerprinting, Zeek, ACME, bcrypt

## Extracurricular

Duke Swim Instructor (25-26), Duke Electrical Vehicles (25-Present), Duke Quantum Information Society Executive Board (25-Present), Duke Applied ML Club (26-Present), Cornell University RL Seminar (26-Present)